# 4.7 The Poisson Random Variable

A random variable X that takes on one of the values 0,1,2,… is said to be a *Poisson* random variable with parameter λ if, for some λ>0,

$$p(i)=P\{X=i\}=e^{-\lambda}\frac{\lambda^i}{i!} \quad i=0,1,2,\ldots \tag{7.1}$$

**Equation (7.1)** defines a probability mass function, since

$$\sum_{i=0}^{\infty} p(i)=e^{-\lambda}\sum_{i=0}^{\infty}\frac{\lambda^i}{i!}=e^{-\lambda}e^{\lambda}=1$$

The Poisson probability distribution was introduced by Siméon Denis Poisson in a book he wrote regarding the application of probability theory to lawsuits, criminal trials, and the like. This book, published in 1837, was entitled *Recherches sur la probabilité des jugements en matière criminelle et en matière civile (Investigations into the Probability of Verdicts in Criminal and Civil Matters)*.

The Poisson random variable has a tremendous range of applications in diverse areas because it may be used as an approximation for a binomial random variable with parameters (n, p) when n is large and p is small enough so that *np* is of moderate size. To see this, suppose that X is a binomial random variable with parameters (n, p), and let λ=np. Then

$$P\{X=i\}=\frac{n!}{(n-i)!i!}p^i(1-p)^{n-i}=\frac{n!}{(n-i)!i!}\left(\frac{\lambda}{n}\right)^i\left(1-\frac{\lambda}{n}\right)^{n-i}=\frac{n(n-1)\cdots(n-i+1)}{n^i}\frac{\lambda^i}{i!}\left(1-\frac{\lambda}{n}\right)^n\left(1-\frac{\lambda}{n}\right)^{-i}$$

Now, for n large and λ moderate,

$$\left(1-\frac{\lambda}{n}\right)^n\approx e^{-\lambda} \quad \frac{n(n-1)\cdots(n-i+1)}{n^i}\approx 1 \quad \left(1-\frac{\lambda}{n}\right)^i\approx 1$$

Hence, for n large and λ moderate,

$$P\{X=i\}\approx e^{-\lambda}\frac{\lambda^i}{i!}$$

In other words, if n independent trials, each of which results in a success with probability p, are performed, then when n is large and p is small enough to make *np* moderate, the number of successes occurring is approximately a Poisson random variable with parameter λ=np. This value λ (which will later be shown to equal the expected number of successes) will usually be determined empirically.

Some examples of random variables that generally obey the Poisson probability law [that is, they obey **Equation (7.1)**] are as follows:

1. The number of misprints on a page (or a group of pages) of a book
2. The number of people in a community who survive to age 100
3. The number of wrong telephone numbers that are dialed in a day
4. The number of packages of dog biscuits sold in a particular store each day
5. The number of customers entering a post office on a given day
6. The number of vacancies occurring during a year in the federal judicial system
7. The number of α-particles discharged in a fixed period of time from some radioactive material

Each of the preceding and numerous other random variables are approximately Poisson for the same reason—namely, because of the Poisson approximation to the binomial. For instance, we can suppose that there is a small probability p that each letter typed on a page will be misprinted. Hence, the number of misprints on a page will be approximately Poisson with λ=np, where n is the number of letters on a page. Similarly, we can suppose that each person in a community has some small probability of reaching age 100. Also, each person entering a store may be thought of as having some small probability of buying a package of dog biscuits, and so on.

**Example 7a**

Suppose that the number of typographical errors on a single page of this book has a Poisson distribution with parameter λ=12. Calculate the probability that there is at least one error on this page.

*Solution*

Letting X denote the number of errors on this page, we have

$$P\{X\geq 1\}=1-P\{X=0\}=1-e^{-1/2}\approx .393$$

**Example 7b**

Suppose that the probability that an item produced by a certain machine will be defective is .1. Find the probability that a sample of 10 items will contain at most 1 defective item.

*Solution*

The desired probability is $(100)(.1)^0(.9)^{10}+(10^1)(.1)^1(.9)^9=.7361$, whereas the Poisson approximation yields the value $e^{-1}+e^{-1}\approx .7358$.

**Example 7c**

Consider an experiment that consists of counting the number of α particles given off in a 1-second interval by 1 gram of radioactive material. If we know from past experience that on the average, 3.2 such α particles are given off, what is a good approximation to the probability that no more than 2 α particles will appear?

*Solution*

If we think of the gram of radioactive material as consisting of a large number n of atoms, each of which has probability of 3.2/n of disintegrating and sending off an α particle during the second considered, then we see that to a very close approximation, the number of α particles given off will be a Poisson random variable with parameter λ=3.2. Hence, the desired probability is

$$P\{ X\leq 2 \}=e-3.2+3.2e-3.2+(3.2)22e-3.2\approx .3799$$

Before computing the expected value and variance of the Poisson random variable with parameter λ, recall that this random variable approximates a binomial random variable with parameters n and p when n is large, p is small, and λ=np. Since such a binomial random variable has expected value np=λ and variance np(1−p)=λ(1−p)≈λ (since p is small), it would seem that both the expected value and the variance of a Poisson random variable would equal its parameter λ. We now verify this result:

$$E[X]=\sum i=0\infty ie-\lambda\lambda ii!=\lambda\sum i=1\infty e-\lambda\lambda i-1(i-1)!=\lambda e-\lambda\sum j=0\infty\lambda jj! \text{ by letting } j=i-1=\lambda \text{ since} \sum j=0\infty\lambda jj!=e\lambda$$

Thus, the expected value of a Poisson random variable X is indeed equal to its parameter λ. To determine its variance, we first compute E[X2]:

$$E[X2]=\sum i=0\infty ie-\lambda\lambda ii!=\lambda\sum i=1\infty e-\lambda\lambda i-1(i-1)!=\lambda\sum j=0\infty(j+1)e-\lambda\lambda jj! \text{ by letting } j=i-1=\lambda\ [\ \sum j=0\infty je-\lambda\lambda jj!+\sum j=0\infty e-\lambda\lambda jj!\ ]$$

where the final equality follows because the first sum is the expected value of a Poisson random variable with parameter λ and the second is the sum of the probabilities of this random variable. Therefore, since we have shown that E[X]=λ, we obtain

$$Var(X)=E[\ X2\ ]-(E[X])2=\lambda$$

Hence, the expected value and variance of a Poisson random variable are both equal to its parameter λ.

We have shown that the Poisson distribution with parameter *np* is a very good approximation to the distribution of the number of successes in n independent trials when each trial has probability p of being a success, provided that n is large and p small. In fact, it remains a good approximation even when the trials are not independent, provided that their dependence is weak. For instance, recall the matching problem (**Example 5m** of **Chapter 2**) in which n men randomly select hats from a set consisting of one hat from each person. From the point of view of the number of men who select their own hat, we may regard the random selection as the result of n trials where we say that trial i is a success if person i selects his own hat, i=1,…,n. Defining the events Ei,i=1,…,n, by

$$Ei=\{ \text{trial i is a success} \}$$

it is easy to see that

$$P\{Ei\}=1n \text{ and } P\{Ei|Ej\}=1n-1,\ j\neq i$$

Thus, we see that although the events Ei,i=1,…,n are not independent, their dependence, for large n, appears to be weak. Because of this, it seems reasonable to expect that the number of successes will approximately have a Poisson distribution with parameter n×1/n=1 and indeed this is verified in **Example 5m** of **Chapter 2**.

For a second illustration of the strength of the Poisson approximation when the trials are weakly dependent, let us consider again the birthday problem presented in **Example 5i** of **Chapter 2**. In this example, we suppose that each of n people is equally likely to have any of the 365 days of the year as his or her birthday, and the problem is to determine the probability that a set of n independent people all have different birthdays. A combinatorial argument was used to determine this probability, which was shown to be less than 12 when n=23.

We can approximate the preceding probability by using the Poisson approximation as follows: Imagine that we have a trial for each of the (n2) pairs of individuals i and j,i≠j, and say that trial i, j is a success if persons i and j have the same birthday. If we let Eij denote the event that trial i, j is a success, then, whereas the (n2) events Eij,1≤i<j≤n, are not independent (see **Theoretical Exercise 4.21**), their dependence appears to be rather weak. (Indeed, these events are even *pairwise independent*, in that any 2 of the events Eij and Ekl are independent—again, see **Theoretical Exercise 4.21**). Since P(Eij)=1/365, it is reasonable to suppose that the number of successes should approximately have a Poisson distribution with mean (n2)/365=n(n−1)/730. Therefore,

$$P\{ \text{ no 2 people have the same birthday } \}=P\{ 0 \text{ successes} \}\approx\exp\{ -n(n-1)730 \}$$

To determine the smallest integer n for which this probability is less than 12, note that

$$\exp\{-n(n-1)730\}\leq 12$$

is equivalent to

$$\exp\{n(n-1)730\}\geq 2$$

Taking logarithms of both sides, we obtain

$$n(n-1) \geq 730 \log 2 \approx 505.997$$

which yields the solution n=23, in agreement with the result of **Example 5i** of **Chapter 2**.

Suppose now that we wanted the probability that among the n people, no 3 of them have their birthday on the same day. Whereas this now becomes a difficult combinatorial problem, it is a simple matter to obtain a good approximation. To begin, imagine that we have a trial for each of the $\binom{n}{3}$ triplets i, j, k, where $1 \leq i < j < k \leq n$, and call the i, j, k trial a success if persons i, j, and k all have their birthday on the same day. As before, we can then conclude that the number of successes is approximately a Poisson random variable with parameter

$$\binom{n}{3} P\{\text{ i,j,k have the same birthday }\} = \binom{n}{3}\left(\frac{1}{365}\right)^2 = \frac{n(n-1)(n-2)}{6 \times (365)^2}$$

Hence,

$$P\{\text{ no 3 have the same birthday }\} \approx \exp\{-n(n-1)(n-2)/799350\}$$

This probability will be less than $\frac{1}{2}$ when n is such that

$$n(n-1)(n-2) \geq 799350 \log 2 \approx 554067.1$$

which is equivalent to $n \geq 84$. Thus, the approximate probability that at least 3 people in a group of size 84 or larger will have the same birthday exceeds $\frac{1}{2}$.

For the number of events to occur to approximately have a Poisson distribution, it is not essential that all the events have the same probability of occurrence, but only that all of these probabilities be small. The following is referred to as the *Poisson paradigm*.

> Poisson paradigm
>
> Consider n events, with $p_i$ equal to the probability that event i occurs, i=1,…,n. If all the $p_i$ are "small" and the trials are either independent or at most "weakly dependent," then the number of these events that occur approximately has a Poisson distribution with mean $\sum_{i=1}^{n} p_i$.

Our next example not only makes use of the Poisson paradigm, but also illustrates a variety of the techniques we have studied so far.

**Example 7d Length of the longest run**

A coin is flipped n times. Assuming that the flips are independent, with each one coming up heads with probability p, what is the probability that there is a string of k consecutive heads?

*Solution*

We will first use the Poisson paradigm to approximate this probability. Now, if for i=1,…,n−k+1, we let $H_i$ denote the event that flips i,i+1,…,i+k−1 all land on heads, then the desired probability is that at least one of the events $H_i$ occur. Because $H_i$ is the event that starting with flip i, the next k flips all land on heads, it follows that $P(H_i)=p^k$. Thus, when $p^k$ is small, we might think that the number of the $H_i$ that occur should have an approximate Poisson distribution. However, such is not the case, because, although the events all have small probabilities, some of their dependencies are too great for the Poisson distribution to be a good approximation. For instance, because the conditional probability that flips 2,…,k+1 are all heads given that flips 1,…,k are all heads is equal to the probability that flip k+1 is a head, it follows that

$$P(H_2|H_1)=p$$

which is far greater than the unconditional probability of $H_2$.

The trick that enables us to use a Poisson approximation is to note that there will be a string of k consecutive heads either if there is such a string that is immediately followed by a tail or if the final k flips all land on heads. Consequently, for i=1,…,n−k, let $E_i$ be the event that flips i,…,i+k−1 are all heads and flip i+k is a tail; also, let $E_{n-k+1}$ be the event that flips n−k+1,…,n are all heads. Note that

$$P(E_i)=p^k(1-p), \; i \leq n-k \qquad P(E_{n-k+1})=p^k$$

Thus, when $p^k$ is small, each of the events $E_i$ has a small probability of occurring. Moreover, for $i \neq j$, if the events $E_i$ and $E_j$ refer to nonoverlapping sequences of flips, then $P(E_i|E_j)=P(E_i)$; if they refer to overlapping sequences, then $P(E_i|E_j)=0$. Hence, in both cases, the conditional probabilities are close to the unconditional ones, indicating that N, the number of the events $E_i$ that occur, should have an approximate Poisson distribution with mean

$$E[N]=\sum_{i=1}^{n-k+1} P(E_i)=(n-k)p^k(1-p)+p^k$$

Because there will not be a run of k heads if (and only if) N=0, the preceding gives

$$P(\text{ no head strings of length } k)=P(N=0) \approx \exp\{-(n-k)p^k(1-p)-p^k\}$$

If we let Ln denote the largest number of consecutive heads in the n flips, then, because Ln will be less than k if (and only if) there are no head strings of length k, the preceding equation can be written as

$$P\{Ln<k\}\approx\exp\{-(n-k)pk(1-p)-pk\}$$

Now, let us suppose that the coin being flipped is fair; that is, suppose that p=1/2. Then the preceding gives

$$P\{Ln<k\}\approx\exp\{-n-k+22k+1\}\approx\exp\{-n2k+1\}$$

where the final approximation supposes that ek−22k+1≈1 (that is, that k−22k+1≈0). Let j=log2 n , and assume that j is an integer. For k=j+i,

$$n2k+1=n2j2i+1=12i+1$$

Consequently,

$$P\{Ln<j+i\}\approx\exp\{-(1/2)i+1\}$$

which implies that

$$P\{\ Ln=j+i\ \}=P\{Ln<j+i+1\}-P\{Ln<j+i\}\approx\exp\{\ -(1/2)i+2\ \}-\exp\{\ -(1/2)i+1\ \}$$

For instance,

$$P\{Ln<j-3\}\approx e-4\approx.0183P\{\ Ln=j-3\ \}\approx e-2-e-4\approx.1170P\{\ Ln=j-2\ \}\approx e-1-e-2\approx.2325P\{\ Ln=j-1\ \}\approx e-1/2-e-1\approx.2387P\{\ Ln=j\ \}\approx e-1/4-e-1/2\approx.1723P\{\ Ln=j+1\ \}\approx e-1/8-e-1/4\approx.1037P\{\ Ln=j+2\ \}\approx e-1/16-e-1/8\approx.0569P\{\ Ln=j+3\ \}\approx e-1/32-e-1/16\approx.0298P\{\ Ln\geq j+4\ \}\approx 1-e-1/32\approx.0308$$

Thus, we observe the rather interesting fact that no matter how large n is, the length of the longest run of heads in a sequence of n flips of a fair coin will be within 2 of log2(n)−1 with a probability approximately equal to .86.

We now derive an exact expression for the probability that there is a string of k consecutive heads when a coin that lands on heads with probability p is flipped n times. With the events Ei,i=1,…,n−k+1, as defined earlier, and with Ln denoting, as before, the length of the longest run of heads,

$$P(Ln\geq k)=P(\text{ there is a string of k consecutive heads })=P(\cup i=1n-k+1Ei)$$

The inclusion–exclusion identity for the probability of a union can be written as

$$P(\cup i=1n-k+1Ei)=\sum r=1n-k+1(-1)r+1\sum i1<\cdots<irP(Ei1\cdots Eir)$$

Let Si denote the set of flip numbers to which the event Ei refers. (So, for instance, S1={1,…,k+1}.) Now, consider one of the r-way intersection probabilities that does not include the event En−k+1. That is, consider P(Ei1⋯Eir) where i1<⋯<ir<n−k+1. On the one hand, if there is any overlap in the sets Si1,…,Sir then this probability is 0. On the other hand, if there is no overlap, then the events Ei1,…,Eir are independent. Therefore,

$$P(Ei1\cdots Eir)=\{\ 0,\quad \text{if there is any overlap in } Si1,\ldots,Sir\ prk(1-p)r,\quad \text{if there is no overlap}$$

We must now determine the number of different choices of i1<⋯<ir<n−k+1 for which there is no overlap in the sets Si1,…,Sir. To do so, note first that each of the Sij, j=1,…,r, refer to k+1 flips, so, without any overlap, they together refer to r(k+1) flips. Now consider any permutation of r identical letters a and of n−r(k+1) identical letters b. Interpret the number of b‘s before the first a as the number of flips before Si1, the number of b‘s between the first and second a as the number of flips between Si1 and Si2, and so on, with the number of b‘s after the final a representing the number of flips after Sir. Because there are (n−rkr) permutations of r letters a and of n−r(k+1) letters b, with every such permutation corresponding (in a one-to-one fashion) to a different nonoverlapping choice, it follows that

$$\sum i1<\ldots<ir<n-k+1P(Ei1\cdots Eir)=(n-rkr)prk(1-p)r$$

We must now consider r-way intersection probabilities of the form

$$P(Ei1\cdots Eir-1En-k+1),$$

where i1<…<ir−1<n−k+1. Now, this probability will equal 0 if there is any overlap in Si1,…,Sir−1,Sn−k; if there is no overlap, then the events of the intersection will be independent, so

$$P(Ei1\cdots Eir-1En-k+1)=[pk(1-p)]r-1pk=pkr(1-p)r-1$$

By a similar argument as before, the number of nonoverlapping sets Si1,…,Sir−1, Sn−k will equal the number of permutations of r−1 letters a (one for each of the sets Si1,…,Sir−1) and of n−(r−1)(k+1)−k=n−rk−(r−1) letters b (one for each of the trials that are not part of any of Si1,…,Sir−1,Sn−k+1). Since there are (n−rkr−1) permutations of r−1 letters a and of n−rk−(r−1) letters b, we have

$$\sum i1<\ldots<ir-1<n-k+1P(Ei1\cdots Eir-1En-k+1)=(n-rkr-1)pkr(1-p)r-1$$

Putting it all together yields the exact expression, namely,

$$P(L_n \geq k) = \sum_{r=1}^{n-k+1} (-1)^{r+1} [(n - rk) + 1 p (n - rk - 1)] p^{kr} (1-p)^r$$

where we utilize the convention that $(m=0j)$ if $m<j$.

From a computational point of view, a more efficient method for computing the desired probability than the use of the preceding identity is to derive a set of recursive equations. To do so, let $A_n$ be the event that there is a string of k consecutive heads in a sequence of n flips, and let $P_n = P(A_n)$. We will derive a set of recursive equations for $P_n$ by conditioning on when the first tail appears. For $j=1,\ldots,k$, let $F_j$ be the event that the first tail appears on flip j, and let H be the event that the first k flips are all heads. Because the events $F_1,\ldots,F_k,H$ are mutually exclusive and exhaustive (that is, exactly one of these events must occur), we have

$$P(A_n) = \sum_{j=1}^{k} P(A_n|F_j) P(F_j) + P(A_n|H) P(H)$$

Now, given that the first tail appears on flip j, where $j<k$, it follows that those j flips are wasted as far as obtaining a string of k heads in a row; thus, the conditional probability of this event is the probability that such a string will occur among the remaining $n-j$ flips. Therefore,

$$P(A_n|F_j) = P_{n-j}$$

Because $P(A_n|H)=1$, the preceding equation gives

$$P_n = P(A_n) = \sum_{j=1}^{k} P_{n-j} P(F_j) + P(H) = \sum_{j=1}^{k} P_{n-j} p j-1 (1-p) + p^k$$

Starting with $P_j=0$, $j<k$, and $P_k=p^k$, we can use the latter formula to recursively compute $P_{k+1}, P_{k+2}$, and so on, up to $P_n$. For instance, suppose we want the probability that there is a run of 2 consecutive heads when a fair coin is flipped 4 times. Then, with $k=2$, we have $P_1=0$, $P_2=(1/2)2$. Because, when $p=1/2$, the recursion becomes

$$P_n = \sum_{j=1}^{k} P_{n-j} (1/2)j + (1/2)k$$

we obtain

$$P_3 = P_2(1/2) + P_1(1/2)2 + (1/2)2 = 3/8$$

and

$$P_4 = P_3(1/2) + P_2(1/2)2 + (1/2)2 = 1/2$$

which is clearly true because there are 8 outcomes that result in a string of 2 consecutive heads: *hhhh, hhht, hhth, hthh, thhh, hhtt, thht*, and *tthh*. Each of these outcomes occurs with probability 1/16.

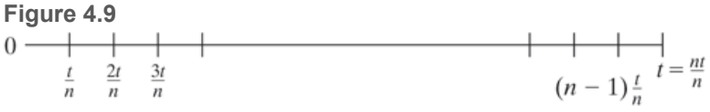
Another use of the Poisson probability distribution arises in situations where “events” occur at certain points in time. One example is to designate the occurrence of an earthquake as an event; another possibility would be for events to correspond to people entering a particular establishment (bank, post office, gas station, and so on); and a third possibility is for an event to occur whenever a war starts. Let us suppose that events are indeed occurring at certain (random) points of time, and let us assume that for some positive constant λ, the following assumptions hold true:

1. The probability that exactly 1 event occurs in a given interval of length h is equal to $\lambda h + o(h)$, where $o(h)$ stands for any function $f(h)$ for which $\lim_{h \to 0} f(h)/h = 0$. [For instance, $f(h)=h^2$ is $o(h)$, whereas $f(h)=h$ is not.]
2. The probability that 2 or more events occur in an interval of length h is equal to $o(h)$.
3. For any integers $n, j_1, j_2, \ldots, j_n$ and any set of n nonoverlapping intervals, if we define $E_i$ to be the event that exactly $j_i$ of the events under consideration occur in the ith of these intervals, then events $E_1, E_2, \ldots, E_n$ are independent.

Loosely put, assumptions 1 and 2 state that for small values of h, the probability that exactly 1 event occurs in an interval of size h equals λh plus something that is small compared with h, whereas the probability that 2 or more events occur is small compared with h. Assumption 3 states that whatever occurs in one interval has no (probability) effect on what will occur in other, nonoverlapping intervals.

We now show that under assumptions 1, 2, and 3, the number of events occurring in any interval of length t is a Poisson random variable with parameter λt. To be precise, let us call the interval [0, t] and denote the number of events occurring in that interval by N(t). To obtain an expression for $P\{N(t)=k\}$, we start by breaking the interval [0, t] into n nonoverlapping subintervals, each of length $t/n$ (**Figure 4.9**).

**Figure 4.9**




Now,

$$P\{N(t)=k\} = P\{k \text{ of the } n \text{ subintervals contain exactly 1 event and the other } n-k \text{ contain 0 events}\} + P\{N(t)=k \text{ and at least 1 subinterval contains 2 or more events}\} \quad (7.2)$$

The preceding equation holds because the event on the left side of **Equation (7.2)**, that is, $\{N(t)=k\}$, is clearly equal to the union of the two

mutually exclusive events on the right side of the equation. Letting A and B denote the two mutually exclusive events on the right side of **Equation (7.2)**, we have

$$P(B) \le P\{\text{ at least one subinterval contains 2 or more events }\} = P(\cup_{i=1}^{n}\{\text{ ith subinterval contains 2 or more events }\}) \le \sum_{i=1}^{n} P\{\text{ ith subinterval contains 2 or more events }\} \text{ by Boole's inequality} = \sum_{i=1}^{n} o(tn) \text{ by assumption 2} = no(tn) = t[\, o(t/n)t/n \,]$$

Now, for any $t, t/n \to 0$ as $n \to \infty$, so $o(t/n)/(t/n) \to 0$ as $n \to \infty$, by the definition of $o(h)$. Hence,

$$P(B) \to 0 \text{ as } n \to \infty \tag{7.3}$$

Moreover, since assumptions 1 and 2 imply that†

$$P\{\text{ 0 events occur in an interval of length } h\} = 1 - [\, \lambda h + o(h) + o(h) \,] = 1 - \lambda h - o(h)$$

†The sum of two functions, both of which are $o(h)$, is also $o(h)$. This is so because if $\lim_{h\to 0} f(h)/h = \lim_{h\to 0} g(h)/h = 0$, then $\lim_{h\to 0}[\, f(h) + g(h) \,]/h = 0$.

we see from the independence assumption (number 3) that

$$P(A) = P\{k \text{ of the subintervals contain exactly 1 event and the other } n-k \text{ contain 0 events}\} = \binom{n}{k}[\, \lambda tn + o(tn) \,]^k [\, 1 - (\lambda tn) - o(tn) \,]^{n-k}$$

However, since

$$n[\lambda tn + o(tn)] = \lambda t + t[o(t/n)t/n] \to \lambda t \text{ as } n \to \infty$$

it follows, by the same argument that verified the Poisson approximation to the binomial, that

$$P(A) \to e^{-\lambda t} \frac{(\lambda t)^k}{k!} \text{ as } n \to \infty \tag{7.4}$$

Thus, from **Equations (7.2)**, **(7.3)**, and **(7.4)**, by letting $n \to \infty$, we obtain

$$P\{N(t) = k\} = e^{-\lambda t} \frac{(\lambda t)^k}{k!} \quad k = 0, 1, \ldots \tag{7.5}$$

Hence, if assumptions 1, 2, and 3 are satisfied, then the number of events occurring in any fixed interval of length t is a Poisson random variable with mean $\lambda t$, and we say that the events occur in accordance with a Poisson process having rate $\lambda$. The value $\lambda$, which can be shown to equal the rate per unit time at which events occur, is a constant that must be empirically determined.

The preceding discussion explains why a Poisson random variable is usually a good approximation for such diverse phenomena as the following:

1. The number of earthquakes occurring during some fixed time span
2. The number of wars per year
3. The number of electrons emitted from a heated cathode during a fixed time period
4. The number of deaths, in a given period of time, of the policyholders of a life insurance company

**Example 7e**

Suppose that earthquakes occur in the western portion of the United States in accordance with assumptions 1, 2, and 3, with $\lambda = 2$ and with 1 week as the unit of time. (That is, earthquakes occur in accordance with the three assumptions at a rate of 2 per week.)

a. Find the probability that at least 3 earthquakes occur during the next 2 weeks.
b. Find the probability distribution of the time, starting from now, until the next earthquake.

*Solution*

a. From **Equation (7.5)**, we have
$$P\{\, N(2) \ge 3 \,\} = 1 - P\{\, N(2) = 0 \,\} - P\{\, N(2) = 1 \,\} - P\{\, N(2) = 2 \,\} = 1 - e^{-4} - 4e^{-4} - 422e^{-4} = 1 - 13e^{-4}$$

b. Let X denote the amount of time (in weeks) until the next earthquake. Because X will be greater than t if and only if no events occur within the next t units of time, we have, from **Equation (7.5)**,
$$P\{X > t\} = P\{N(t) = 0\} = e^{-\lambda t}$$

so the probability distribution function F of the random variable X is given by
$$F(t) = P\{\, X \le t \,\} = 1 - P\{X > t\} = 1 - e^{-\lambda t} = 1 - e^{-2t}$$

## 4.7.1 Computing the Poisson Distribution Function

If X is Poisson with parameter $\lambda$, then

$$P\{X=i+1\}P\{X=i\}=e-\lambda\lambda i+1/(i+1)!e-\lambda\lambda i/i!=\lambda i+1 \tag{7.6}$$

Starting with $P\{X=0\}=e-\lambda$, we can use (**7.6**) to compute successively

$$P\{\ X=1\ \}=\lambda P\{\ X=0\ \}P\{\ X=2\ \}=\lambda 2P\{\ X=1\ \}\vdots P\{\ X=i+1\ \}=\lambda i+1P\{\ X=i\ \}$$

We can use a module to compute the Poisson probabilities for **Equation (7.6)**.

***Example 7f***

a. Determine $P\{X\leq 90\}$ when X is Poisson with mean 100.
b. Determine $P\{Y\leq 1075\}$ when Y is Poisson with mean 1000.

Solution

Using the Poisson calculator of StatCrunch yields the solutions:

a. $P\{X\leq 90\}=.17138$
b. $P\{Y\leq 1075\}=.99095$